

Towards a Bottom-Up Development of Reference Architectures for Smart Energy Systems

Maximilian Irlbeck*, Denis Bytschkow**, Georg Hackenberg*, and Vasileios Koutsoumpas*

* Fakultät für Informatik, Technische Universität München, Munich, Germany

** fortiss GmbH, Munich, Germany

E-Mail: {irlbeck,hackenberg,koutsoumpas}@in.tum.de, bytschkow@fortiss.org

Abstract—Smart energy systems seem a promising choice for countries worldwide to realign their power systems to the challenges predicted for the next decades. With the will to participate in this class of systems, many solution providers design custom systems, which sometimes consist of similar parts, but are on the contrary hard to compare to each other. However, a reference describing existing commonalities is needed as a basis for many activities such as regulation design, legislation, national discussion or standardization. This paper illustrates the challenges connected with the creation of reference architectures for smart energy systems, delineates their benefits and suggests a model and method for their incremental, bottom-up development and validation through concrete system architectures.

Index Terms—Reference Architecture; Smart Energy Systems; Smart Grid; Ontologies; Conceptual Modeling; Domain Modeling; Bottom-up design;

I. INTRODUCTION

The existing energy systems will go through major changes within the next decades. Facts like the increasing of renewable, decentralized energy sources, the growing number of electric cars, national efforts on market liberalization and reduction of CO_2 -emissions, integration of different energy grid types (e.g. electric power, district heating or gas grids) or the need for advanced monitoring systems and increased power stability will drive this change. Many countries discuss new concepts to solve problems like the fluctuating supply of renewable energy with smart systems while having to ensure power stability.

As a reaction to this worldwide trend, many actors think of a new class of systems, often labeled as "Smart Energy Systems" (SES) and develop new systems inside this class. However these systems focus on different aspects of the energy system, involve different stakeholders, include new components, functions and data structures and use different technologies, concepts, terminology and infrastructure. The class of SES comprises a variety of systems used in home appliances, energy management, district heating, intelligent devices, virtual power plants, demand side management, market places, data platforms, metering infrastructure, field devices, portal software, weather forecasting or grid operations.

Many countries, national and international organizations are interested in SES, as this class of systems is expected to have an impact on national grid infrastructure, markets, customers and industries. In contrast the sheer amount of existing systems and their different architectures complicate the comprehension and comparison of different solutions or the elaboration of an abstract view on SES.

Problem Statement: For many tasks that need an abstract system comprehension such as the design of incentive-based subsidies or regulatory frameworks, the development of standards, legislation, discussions on a national or international level, market design or the collaboration between different domains, a reference architecture is of great benefit. Consequently, the validity and feasibility of a reference architecture has to be shown for the class of systems, otherwise its chance of acceptance is low. Existing reference architectures do not provide an explicit representation of existent systems and are difficult to develop and to validate.

Research Objective: We believe that models will be a valuable help in the complex design of reference architectures. Therefore our goal is to shape and elaborate an approach that bases on existent systems and employs dedicated models and a consistent methodology for the creation of reference architectures. These results facilitate the analysis of existing smart energy systems as well as the creation of abstract view on SES, which is valuable for various tasks.

Contribution: This paper focuses on reference architectures and their bottom-up development. It illustrates challenges and benefits, confronts our understanding of the term with other popular definitions, presents other existing approaches and formulates general requirements. Finally the article presents an approach for the incremental development of reference architectures for smart energy systems and evaluates its application with a case study.

II. MOTIVATION

A. Definition: What is a reference architecture?

There is no widely accepted definition of the term *reference architecture*. Therefore we want to first delineate our understanding and compare it to other definitions.

In our point of view, a *reference architecture* is a reference model that captures architectural knowledge about a class of systems; based on individual system architectures it encompasses central architectural concepts of the class and their interrelation. It must be supported by a unified, explicit, unambiguous, and widely understood domain terminology.

Other definitions take a different view on the term. Kruchten [1] provides a process-centric definition of the term as part of the Rational Unified Process (RUP). Within RUP reference architectures can be seen as architectural patterns for a particular context that are designed for reuse and derived

from best practice. In contrast, Bass et al. [2] gives a model-centric definition of the term as part of their work on software architectures in general. Their reference architecture model comprises the functionality of the system, the required data flows as well as the mapping to concrete software implementations. Consequently, Cloutier et al. [3] coins a knowledge-centric definition of the term as part of their study on common reference architecture usage. The body of knowledge ranges from high-level business goals such as market segmentation to low-level design rules such as essential design patterns. Estefan et al. [4] highlight the implementation-independent character of the term and explain its difference to other reference models. They argue that reference architectures are not required to be concrete architectures and can be defined at many levels of detail or abstraction and used for many different purposes. Finally, Nakagawa et al. [5] note the difference between reference architectures and product line architectures, which is commonly overseen. According to the authors product line architectures are more specialized for a family of products, while reference architectures cover a more general domain view.

B. Benefits: Why do we need RAs?

In general, reference architectures enable the discussion about a class of systems including widely accepted architectural properties. Consequently, a common language can be used rather than system-dependent terms and concepts, which eases the knowledge exchange between stakeholders. On the contrary, the absence of an explicit reference architecture might lead to misunderstandings among people from different organizational backgrounds. Reasons for misunderstandings are for example (1) concepts that sound similar but are essentially different in purpose or (2) concepts that sound different but are essentially similar in purpose. Aside of improved knowledge exchange, reference architectures further enable the comparison between individual systems with respect to reference concepts. Comparison is particularly helpful in case national regulations, laws or standards have to be derived from pilot projects such that a wide range of vendor products and services is supported effectively. In addition to knowledge exchange and system comparison, reference architectures enable the reuse of contained concepts such as software components in case they fulfill the requirements of the application. Consequently, new components do not need to be designed from scratch, but the reference architecture offers access to a whole body of experience from previous systems. Reuse finally eases the design and integration of new components ensuring higher quality and interoperability.

C. Challenges: Is it really so hard to create RAs?

Multi-disciplinary systems involve a high degree of interaction between people, software and hardware; hence, they are intrinsically complex. Because of this complexity, the design of such systems belong to a class of problems known as *wicked problems*, which was coined by Rittel and Webber [6].

The clash of multiple disciplines, different sectors, numerous enterprises and organizations with own goals and visions complicate the elaboration of a common view and complicate engineering tasks [7]. Moreover, impact factors such as existing laws, national agendas, standards, market and regulation policies have to be taken into consideration. The mixture of different maturity levels of smart energy systems from sketchy system concepts to mature, industrial-strength solutions increase the complexity further. Thus, the development of a reference architecture, where every of the above mentioned aspects influences the system design is a hard task.

Ramesh and Jarke[8] point out that not every domain is sufficiently standardized to establish a reference architecture that provides a blueprint of final products. Reference architectures are in their view an abstraction of best practice, condensed from numerous case studies over an extended period of time, followed by more case studies to refine and evaluate the proposed reference model. There is nothing provably correct about reference architectures; they derive their relevance from the slice of practice they cover.

III. RELATED WORK

A. Reference Architectures

Common system understanding and a clear terminology is of paramount importance to develop large systems including many stakeholders and different disciplines. Many researchers have shown the significance of reference architecture and the role that it plays for system understanding. Mature domains, such as compilers, databases and operating systems, have well-known reference architectures [9], [10], which enables modular development, re-use of components and their integration into the system, leading to lower costs and better quality. More formal modeling, designing, specifying, tracing and refinement for a specific domain have been shown in [11], [12], [13]. Our approach focuses on establishing a reference architecture for the smart energy domain. In contrast to pure software systems with a clear scope smart energy systems requires the combination of many diverse aspects, such as the underlying physical system, which needs to be controlled with software components, planning and financial clearance as well as goals of many different stakeholders, such as private house owners, companies, public organizations, system operators, politicians, certification agencies and policy makers. Furthermore the structure of ownership and access to data is not clear, which exacerbates the creation of successful reference architecture.

Important factors for a successful reference architecture are provided by the classification of Angelov et al. [14]. Here, five main types of reference architectures are defined according the dimensions of context, goals and design. According to Angelov, reference architectures that can be classified in one of these types have better chances to become a success.

B. Methods for Reference Architectures

With respect to existing methods for developing reference architectures for multi-disciplinary systems there is little lit-

erature available. Most approaches concentrate on a specific domain and are based on empirical studies. Ramesh and Jarke [8] point out that the traditional normative computer science approach of imposing such models on developers is long known to have failed in most cases. Therefore, they propose a method based on empirical studies for developing reference models for requirements traceability. Focus groups and interviews conducted in 26 major software development organizations demonstrate a wide range of traceability practices with distinct low-end and high-end users of traceability. From these observations, reference models comprising the most important kinds of traceability links for various development tasks have been synthesized. This approach points out not only how to develop reference models for requirements traceability but also how to set up such an empirical study for developing reference models.

Nakagawa et al. [15] identify a lack of work that investigates the essence of reference architectures, their dimensions and elements that they should contain. Thus, their main contribution is a reference model for reference architectures, named RAModel (Reference Architecture Model) that intends to improve the understanding about what reference architectures are, as well as their components and relationships, supporting the establishment, use, and evolution of such architectures. The method introduced is divided in three basic steps: 1) Identification of the information sources 2) Identification of elements that could be contained in RA and 3) Design of the reference model for RA.

The IFIP/IFAC Task Force on Architectures for Enterprise Integration develop in their work the Generic Enterprise Reference Architecture and Methodology (GERAM) [16]. The GERAM framework has an integrated approach in order to organize the existing enterprise knowledge and facilitate creation and maintenance of enterprises for their entire life-history. The IFIP/IFAC Task Force combines in this approach product models which business process design patterns and offers different views on the system. Although, GERAM offers an extensive framework, it remains on a very abstract level providing mainly the scope of the system without clear and explicit options for refinement steps [17]. This makes it rather difficult to use this approach for formalization and mapping of systems to new reference architecture, especially within a domain such as smart grids.

C. Reference Architectures for Smart Grids

In 2011 the international standard *IEEE P2030 Draft Guide for Smart Grid Interoperability* was published. The guide aims to provide a common understanding, terminology, definitions and guidance for design and implementation of Smart Grid components and end-use applications for both legacy and future infrastructures [18]. P2030 offers three major perspectives, the power systems perspective, communication technology perspective and the information technology perspective. Furthermore, each perspective is comprised of seven domains: bulk generation, transmission, distribution, service providers, markets, control/ operations and customers. Every domain

consists of a number of entities, which are logically connected with interfaces. IEEE P2030 seems to be promising reference architecture for standardization of interfaces, however, there is no evidence how well existing smart grid concepts can be represented in this approach, since links between different perspectives are not presented. Therefore it remains unclear if this approach is feasible for rigorous analysis with respect to functionalities, envisioned for the smart grid, and their manifestation within the system.

European Smart Grid activities are working on other reference architectures. The most promising candidate seems to be the Mandate M/490, where the CEN-CENELEC-ETSI Smart Grid Coordination Group is requested to provide a reference Smart Grid Architecture Model (SGAM) framework [19]. Instead of perspectives and domains SGAM uses a three dimensional structure with: five interoperability layers representing different abstraction views - business, function, information, communication and component; five domains representing the physical energy value chain levels - generation, transmission, distribution, distributed energy resources (DER) and customer premises; five zones representing the power management levels - process, field, station, enterprise, market. Furthermore, SGAM mentions cross cutting issues, such as security. The reference architecture has three major objectives: to represent the main smart grid elements with all major stakeholders, to support a variety of different approaches and to provide the basis to search for standards or identification of possible lacks. Use cases are used for the mapping process. The SGAM method starts on the bottom layer for components and continues on the business layer to include organizations and stakeholders. Afterwards the mapping is carried out from business layer down to communication layer. An explicit interconnection between different layers is missing preventing analysis capabilities over several layers.

Appelrath et al. [20] presents a software architecture as basis for the development of new Information and Communication Technology (ICT) architectures for the future energy system. The introduced method and the presented Electricity Market Reference Architecture (ERA) have its origin in the German research project eTelligence, where a SES was designed and implemented, although ERA was not used as reference architecture for this purpose. ERA relies on existing conceptional approaches such as IntelliGrid [21]. It describes the interplay of different market roles in an abstract way, independent of the selected technologies for their implementation. The architectural view of ERA is oriented to four layers proposed by the Open Group Architecture Frameworks (TOGAF): business-, data-, information- and technology layer. Finally, with respect to the technical interpretation of the developed reference architecture, UML Component diagrams are used to represent at each layer the corresponding results. These diagrams are enhanced with informal notions of possible instantiation of components and representation of constraints as UML Comments. The lack of formal semantics, the not existent proof of validity or reference character and the missing

link to the transferability to concrete systems complicate the use of ERA as reference architecture.

Wietfeld et al. [22] introduce a reference architecture design offering a set of extensibility points to existing solutions for increased flexibility in moving energy markets. The work concentrates mainly on the design of an ICT reference architecture for regional energy marketplaces. It considers requirements for ecologically as well as economically sustainable energy supplies. The work was conducted as part of the E-DeMa research project. However, there is no clear understanding how the presented RA can be adapted to other SES.

IV. APPROACH

Our experience is based on the evaluation of different system architectures within the German smart grid project *E-Energy*¹, in which six model regions develop individual smart grid solutions.

We delineate in the following section general requirements and present a structured approach that supports the creation of reference architectures. This approach is currently applied in E-Energy to derive a reference architecture out of different system architectures.

A. General Requirements

The definition given in II-A illustrates, that architectures of a set of SES are the basis for a reference architecture. This set of systems is called *systems under consideration*. Their architectures are used as the starting point, the justification and validation for the reference architecture.

The selection and interpretation of the systems under consideration as well as the creation of the reference architecture is conducted by a group of experts in the field of SES, namely *reference architects*.

The systems under consideration share many similar concepts in their architectures - concepts like components, functions, data structures, processes, use cases and their interplay. A subset of those concepts is the basis for the reference architecture. Since all concepts are developed in independent projects, they differ in their:

- 1) naming of concepts & terminology,
- 2) architectural description techniques,
- 3) level of detail & completeness,
- 4) technical scope & design.

1) *Different terminology*: Architectures do not consist only out of building blocks and their interrelation. They involve an often implicitly defined terminology. Each part of the architecture has a name. This name identifies abstract parts of the architecture, has a meaning, which may be indicated by the naming, is used as a reference in different parts of the architecture, but lacks often an explicit definition. In informal architecture descriptions many names are used for the same concept, without explicitly stating that two names are used synonymously.

Different systems may share terms, especially when they belong to the same class of systems. Nevertheless the meaning of these terms in the context of a class of systems is ambiguous and not defined, as there is no frame of reference.

For SES this is indeed a problem, because it is not clearly defined what terms like demand side management, load shaping or energy management or virtual power plants actually mean. The decision to what extent a virtual power plant may include consumers, how it is organized or in which way it is discriminable from the concept of an aggregator is not clear. Moreover the interrelation of terms is not defined, which hinders a systematic understanding of several terms. However these terms are used to discuss SES. This situation can lead to unnoticed misunderstandings and inefficient communication.

Some terms have to be newly created to identify concepts on the level of a reference architecture. Their interrelation to existing terms on a system level has to be shown to ensure an explicit mapping on existent systems.

To this end a model for reference architectures has to address terminological aspects, describing terms in use in context of systems and on a reference level and their respective meaning. It has to provide mechanisms to define terms, to interrelate them, to link them to architectural concepts and to express linguistic characteristics of e.g. synonymy.

2) *Different architectural description techniques*: Natural language representation, "power-point" shape diagrams, UML based models and diagrams or architecture description languages [23] are common ways to describe a system's architecture. Unfortunately these different descriptions complicate the analysis of system architecture and therefore the creation of a reference architecture.

A uniform representation can avoid this problem by representing concepts of different systems in the same way. This representation permits the uniform description of relevant concepts on the reference and system level and the mapping between both levels.

To cover a wide variety of architectural descriptions the representation should include a set of basic architectural concepts (e.g. components, functions, data structures or interfaces) and should address their flexible connection.

3) *Different levels of detail, completeness & preciseness*: Many architectural documentations differ in their level of detail, completeness and preciseness. While detailed descriptions offer too many details, high level descriptions as well as not completely accessible descriptions may be too vague.

Within the analysis new architectural knowledge may become apparent and has to be included. Therefore the model has to offer flexible ways to use abstraction in order to reduce complexity as well as refinement to include new details. It has to be capable to express imprecise knowledge and its precision in the course of the analysis.

4) *Different design of systems*: There are many ways systems can be designed to solve a certain problem. As the energy system consists of many parts, there is room for many solutions. The variety of system designs for a specific problem

¹cf. <http://www.e-energy.de/en/>

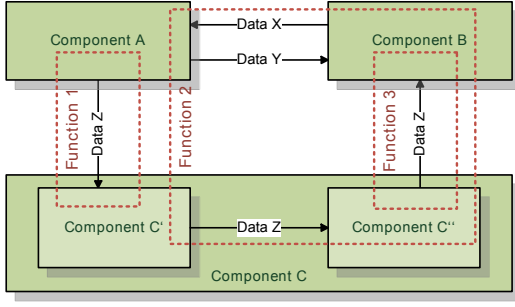


Fig. 1. A model for a common architecture description: Components and subcomponents are connected through channels, which enable the exchange of specific data. Functions and encapsulated subfunctions involve components and channels/data.

is tremendous. Thus the resulting architectures differ in their structure and seem to be incomparable.

Nevertheless there exists a level of abstraction which is suitable to compare different solutions and to show conceptual similarity of different designs. We see abstraction and refinement as the central concepts to reach this level of abstraction. This must be addressed by the modeling technique applied.

B. Method

The following section illustrates the method we propose for the modeling of reference architectures. The method was developed for the evaluation of the six model regions and their ICT systems inside the German smart grid project *E-Energy* under the mandate of the ministry of economics and technology.

Thus the proposed model focuses on the special needs of legislation and regulatory design, with their need to identify parts of the system, to assure which data is stored and transferred and to understand the functions of each part in the system.

To represent system architectures, we use a common description of architectures, which is shown by Fig. 1. A system is decomposed into components, which themselves can be further refined in components. Components are building blocks that encapsulate a subset of the systems functionality and/or data and restrict access to them via explicit interfaces [23].

Typed channels between components allow the directed exchange of data. Functions fulfill a specific task in the system and can involve a set of components, channels and data. They can be further refined to subfunctions, which encapsulate more specific tasks, that are part of the original function. Data structures contain bits of information, which can be stored by components or used by operations inside functions.

To formalize the information which resides in this model we choose the modeling paradigm of ontologies. Ontologies offer a formal, explicit specification of a shared conceptualization [24].

With respect to the architectural knowledge we want to represent, we use an intentionally simple meta model, which is shown in Fig. 2. It is able to formalize knowledge about architectures and uses three ontological classes (*Component*, *Func-*

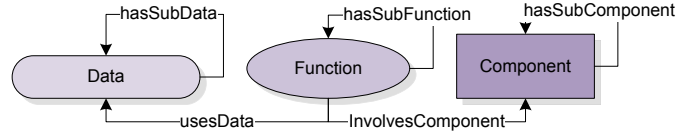


Fig. 2. A simple ontological model for the representation of architectural knowledge.

tion, *Data*) and relations (*hasSubData*, *usesData*, *hasSubFunction*, *involvesComponent*, *hasSubComponent*) between them.

This model can be used to describe knowledge about individual systems as well as to represent the reference architecture. Additionally, it uses the relation *isA*, also known as *subclassOf*, to represent inheritance between concepts and to map the reference architecture to the system level.

We want to emphasize that we could apply other modeling techniques which may fulfill our requirements. The presented model describes the essence of what we need in order to reach our goal. If another modeling technique is able to match the model, it could easily fit into place.

We structure our model in three levels: *query level*, *reference level* and *system level*.

The **query level** offers a user of the model an uniform interface to the reference architecture in the form of queries. Although not described in this paper, it uses the model to create architectural views out of queries, e.g. a top level black box view hiding internal functions or a data centric view showing which data is distributed on which components. The query level addresses stakeholder specific needs, eases the access to the reference level and increases the tangibility of the reference architecture.

On the **reference level** the concepts of reference architecture are modeled. These concepts represent an explicit conceptualization of the system class and connections to the systems under consideration via *subclassOf* relations. The connection allows tracing of each concept to a system specific concept.

Finally, the **system level** represents the architectural knowledge of each of the systems under consideration.

C. Procedure

In the following we propose the general steps of a procedure for the creation of reference architectures, based on the presented model and requirements. The procedure consists of seven steps and can be repeated iteratively to refine the resulting reference architecture. Each iteration can cover a different aspect of the class of systems. The presented steps are eventually not consecutive in reality because some actions overlap.

Step 1. Selection of the systems under consideration

This step involves the analysis of available systems within the class of systems, the estimation of their relevance and the acquisition of their architectural documentation. This step is crucial for the creation of reference architectures (cf. Section IV-A). The result of this step is a *set of systems under consideration*.

Step 2. Extraction of architectural knowledge

The analysis of the available architectural documentation of each system is a time consuming task. The goal is to

identify central concepts of the system, their rationale and architectural role using the available documentation and by integrating system experts. The result of this step is a *identification of central architectural concepts* of each system.

Step 3. Creation & validation of the system model

The task of this step is to model the architectural knowledge in a formal and explicit way. This step ensures the uniform representation of each system's architecture. After that, experts can use the model to check its conformance to the system architecture. The result of this step are *validated models of the systems under consideration*.

Step 4. Identification and renaming of relevant concepts

The comparison of concepts on the system level and the check of their relevance for the reference architecture is the purpose of this step. Within each system and across different systems relevant concepts have to be identified and represented on the reference level. Whenever differences in system design hinder a representation on the reference level, the choice of an appropriate level of abstraction is very important, to equalize differences in system design. The use of existing terms (stemming e.g. from standardized glossaries) or the creation of new reference terms circumvents the problem of different naming in systems. The result of this step is a *set of reference concepts*.

Step 5. Modeling of the reference level

The basis of this step are the identified reference concepts. They are interrelated to model the reference architecture as a representation of the central architectural aspects of the class of systems. The result of this step is a *model of the reference architecture*.

Step 6. Mapping on the system level

In this step each reference concept is mapped on the matching concepts on the system level using appropriate relations (e.g. *subclassOf*). This translation of reference concepts to the system level enables the better comparison of individual systems as well as the improved accessibility to individual systems. The result of this step is the *interrelation of the reference level with the system level*.

Step 7. Validation of the resulting reference architecture

The creation of reference concepts and their mapping to system level concepts may be invalid. Therefore the task in this step is to present the resulting model to system experts and users of the RA in an appropriate form. They have to check whether their system is represented correctly. The result of this step is a *validated reference architecture*.

V. CASE STUDY

In the following we want to illustrate how our approach can be used to model reference architectures. Therefore we introduce a simple example of an SES and show the abilities of our approach. Based on examples from the E-Energy project and smart energy research demonstrators [25] we derive a case study with two typical examples *Smart Butler* and *E-Manager*, which are used as study objects in the following. We follow the procedure presented in section IV-C.

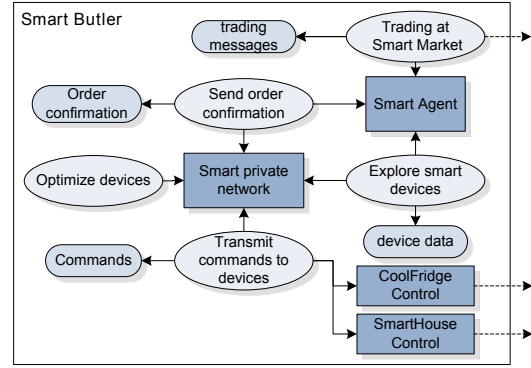


Fig. 3. The model of the architectural knowledge of the Smart Butler system.

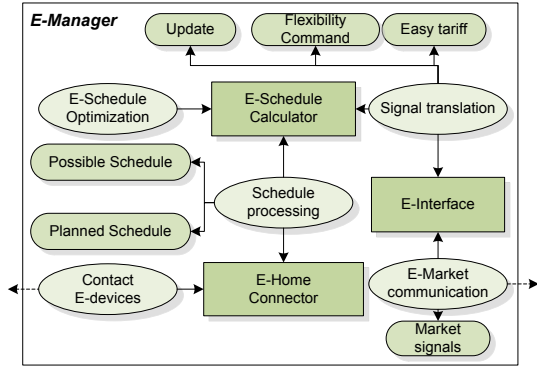


Fig. 4. The model of the architectural knowledge of the E-Manager system.

Step 1 + 2: Our study objects are located within the customer segment. The goal of these systems is to control several devices in the house and adjust their energetic behavior within predefined limits to reach desired goals like energy saving, load shifting, comfort functions or partly self-sustaining supply. The respective architectural documentation was analyzed thoroughly in this step to gain architectural knowledge.

Step 3: The models of the architectural knowledge of each system can be found in Fig. 3 (*Smart Butler*) and Fig. 4 (*E-Manager*). The models show the different naming and design of both systems, though their goals are similar.

In the *Smart Butler* system the function *Trading at Smart Market* gives the *Smart Agent* the possibility to bid on an external market, called *Smart Market*. He is able to buy energy for a certain block of time or he can offer services like to provide the flexibility of devices for energy consumption. The *Smart private network* connects the *Smart Agent* with the home devices via the function *Explore smart devices*. He receives the characteristic of the individual devices which are registered at the *Smart private network*. The *Smart Agent* is able to accept offers and sends the order confirmation to the *Smart private network* which tries to optimize several devices based on this confirmation (function *Optimize devices for confirmations*). It is able to send signals to several devices through the use of the function *transmit signals to devices* and two components which encapsulate the control of the devices.

The *E-Manager* system uses the *E-Interface* to communicate with the E-Market. The component receives and sends so called *Market signals*. If a signal is received it is translated by the E-Interface to other data called *Easy tariff*, *Flexibility command* and *Update* and sent to the *E-Schedule Calculator*. Depending on the signal type, the *E-Schedule Calculator* can optimize the schedules of devices according to a price curve (*Easy tariff*), according to a grid bottleneck (*flexibility command*). The *E-Schedule Calculator* uses the function *Schedule processing* in two ways. On the one hand it is able to receive possible schedules from the devices (through the use of the *E-Home Connector* and the function *Contact E-Devices*), which contain potential load shifting potentials. On the other hand it can use the function to transmit optimized schedules as *planned schedules* to the devices. The signal *Update* plays a special role. It is a command of the E-Market to an E-Manager to transmit its aggregated possible schedule for reasons of grid forecasts. This is achieved through the functions *Contact E-devices*, *Schedule processing*, *Message translation* and *E-Market Communication*.

Step 4: This step is crucial for the derivation of a valid reference architecture. The relevant concepts have to be identified across many systems - in our case by only comparing two systems. The decision which concepts are considered as relevant for the reference architecture is influenced by several factors which also reside outside the system. National regulation, customer needs or the ability of concepts to stabilize the power grid may be such factors.

In both systems we see three logical parts that are similar from a more abstract view that ignores irrelevant details of both systems (e.g. functions like *Signal translation* or components like *CoolFridge control*). Those similar parts are first the communication with market places, second the communication with devices and finally the optimization of the devices. Therefore we rename system specific terms for components by introducing the reference terms *Gateway*, *Device Abstraction* and *Local Optimization*. We structure and rename the functions and data accordingly to *Optimize device schedule*, *Communication with devices*, *Market communication* and *Market message* and its respective subclasses.

Step 5 + 6: We connect the relevant concepts of the reference architecture and map them to their system specific counterparts. The resulting reference is shown by Fig. 5.

In step 7 the elaborated reference architecture has to be validated by system experts. In this case study we have to skip this step.

VI. DISCUSSION

In summary we have presented a method for the creation of a reference architecture which is based on architectures of systems of different maturity. This is especially important for smart grids and the smart energy systems sector.

Architectural knowledge of existing systems, pilots, research projects or use cases are included into models and used as a basis for reference architectures. This property distinguishes our approach from other similar approaches and

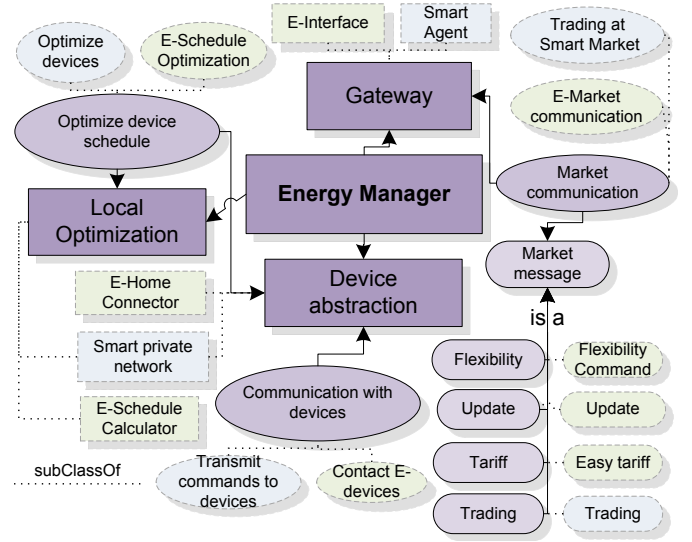


Fig. 5. The concepts of the reference architecture (violet) and their mapping to system specific concepts.

existing reference architectures in general, which are mostly defined without explicit reference to existing systems. It allows an inherent mapping of systems to the reference architecture irrespective of the way the system is described.

The meta model we apply is intentionally kept simple in order to reduce the effort to create a reference architecture as well as to find a common conceptualization for many systems.

The method we presented is based on our experience from the evaluation of six Smart Grid model regions within the E-Energy projects and is feasible to capture essential architectural knowledge. Various architectures of systems using different representation, designs and terminology can be easily mapped to this structure. Consequently, the model can be used and extended by non-experts, which is very important for discussions in a multi-disciplinary domain, such as the smart grid.

Additionally, the model provides enough formality to analyze system properties, identify important interfaces for standardization and define certification criteria.

Certification discussions in Germany were facing an area of conflict for smart meter devices. On the one hand, smart meter should be able to provide enough data for system operators, on the other hand, legal requirements claim for strict communication limitations because of privacy issues. This enforces smart meter providers to build two separate modules leading to higher prices for consumers and fuzzy regulations for the data. In contrast to the IEEE P2030 and the SGAM architecture, our approach intrinsically grasps such problems and allows a structured analysis and discussions about such architectural issues.

As a downside our method lacks of the completeness other reference architectures offer. It is not able to describe e.g. rationale behind architectural designs, business rules or technical details.

The modeling paradigm of ontologies matches our requirements of architectural knowledge representation, because

ontologies offer a way to represent shared knowledge about a subject of interest. Their ability to render shared vocabulary of a domain and relationship of concepts [26], enables us to explicitly describe terms and their meanings. Additionally the ability of ontologies to represent an explicit representation of a shared conceptualization makes them good choice to model reference architectures.

Currently, the meta model is suitable to represent all relevant architectural knowledge we obtained so far within the E-Energy project. However, we think of extensions by other elements such as processes, which represent arrangement of functions providing an additional detail that might be important.

This paper showed the application of our method only for a small case study to demonstrate the basic ideas. A large case study with this method is currently conducted in the E-Energy project to validate this approach and to gain further insights.

VII. CONCLUSION & FUTURE WORK

Reference architectures are essential for the comprehension of complex systems such as the smart grid. Their influence on the discussions of future energy systems is tremendous, as they offer a general view on a complex subject many people can understand.

We demonstrated in this paper the basic ideas of a bottom-up approach for the development of reference architectures. It encompasses the architectural knowledge of individual systems to create a common vision where each of these systems is represented. The reference level offers a general understanding of architectural terms and their meaning. The traceability from the reference level to the system level eases the comprehension of individual systems as well as the comparison of different smart energy systems.

The case study offered promising results, but the approach has to show its applicability and scalability to larger sets of systems. A first step in that direction is the evaluation of the E-Energy systems. Open issues like the elaboration of a more detailed procedure, the integration of further modeling elements, the development of views on the reference architecture, the evaluation of other modeling paradigms, the proof of the usefulness of our approach and the detailed comparison with other reference architectures will drive our future research.

ACKNOWLEDGMENT

The authors address special thanks to the consortium of the project *E-Energy* and to the *German Ministry of Economics and Technology*.

REFERENCES

- [1] P. Kruchten, *The rational unified process: an introduction*. Addison-Wesley Professional, 2004.
- [2] L. Bass, P. Clements, and R. Kazman, *Software architecture in practice*. Addison-Wesley Professional, 2003.
- [3] R. Cloutier, G. Muller, D. Verma, R. Nilchiani, E. Hole, and M. Bone, "The concept of reference architectures," *Systems Engineering*, vol. 13, no. 1, pp. 14–27, 2010.
- [4] OASIS, *Reference Architecture Foundation for Service Oriented Architecture Version 1.0 (Committee Specification 01)*, Organization for the Advancement of Structured Information Standards, Dec. 2012.
- [5] E. Nakagawa, P. Oliveira Antonino, and M. Becker, "Reference architecture and product line architecture: A subtle but critical difference," *Software Architecture*, pp. 207–211, 2011.
- [6] H. Rittel and M. Webber, "Wicked problems," *Man-made Futures*, pp. 272–280, 1974.
- [7] G. Hackenberg, M. Irlbeck, V. Koutsoumpas, and D. Bytschkow, "Applying formal software engineering techniques to smart grids," in *Software Engineering for the Smart Grid (SE4SG), 2012 International Workshop on*. IEEE, 2012, pp. 50–56.
- [8] B. Ramesh and M. Jarke, "Toward reference models for requirements traceability," *Software Engineering, IEEE Transactions on*, vol. 27, no. 1, pp. 58–93, 2001.
- [9] A. Sheth and J. Larson, "Federated database systems for managing distributed, heterogeneous, and autonomous databases," *ACM Computing Surveys (CSUR)*, vol. 22, no. 3, pp. 183–236, 1990.
- [10] A. Hassan and R. Holt, "A reference architecture for web servers," in *Reverse Engineering, 2000. Proceedings. Seventh Working Conference on*. IEEE, 2000, pp. 150–159.
- [11] K. Aberer, L. Alima, A. Ghodsi, S. Girdzijauskas, S. Haridi, and M. Hauswirth, "The essence of p2p: A reference architecture for overlay networks," in *Peer-to-Peer Computing, 2005. P2P 2005. Fifth IEEE International Conference on*. IEEE, 2005, pp. 11–20.
- [12] C. Lin, S. Lu, X. Fei, A. Chebotko, D. Pai, Z. Lai, F. Fotouhi, and J. Hua, "A reference architecture for scientific workflow management systems and the view soa solution," *Services Computing, IEEE Transactions on*, vol. 2, no. 1, pp. 79–92, 2009.
- [13] D. Weyns and T. Holvoet, "A reference architecture for situated multiagent systems," *Environments for Multi-Agent Systems III*, pp. 1–40, 2007.
- [14] S. Angelov, P. Grefen, and D. Greefhorst, "A classification of software reference architectures: Analyzing their success and effectiveness," in *Software Architecture, 2009 & European Conference on Software Architecture. WICSA/ECSA 2009. Joint Working IEEE/IFIP Conference on*. IEEE, 2009, pp. 141–150.
- [15] E. Nakagawa, F. Oquendo, and M. Becker, "Ramodel: A reference model for reference architectures," in *Software Architecture (WICSA) and European Conference on Software Architecture (ECSA), 2012 Joint Working IEEE/IFIP Conference on*. IEEE, 2012, pp. 297–301.
- [16] I. T. Force, "Geram: Generalised enterprise reference architecture and methodology," *version*, vol. 1, no. 2, p. 30, 1998.
- [17] S. Aier, C. Riege, and R. Winter, "Unternehmensarchitektur–literaturüberblick und stand der praxis," *Wirtschaftsinformatik*, vol. 50, no. 4, pp. 292–304, 2008.
- [18] IEEE, *IEEE Guide for Smart Grid Interoperability of Energy Technology and Information Technology Operation with the Electric Power System (EPS), End-Use Applications, and Loads*, Institute of Electrical and Electronics Engineers, Sep. 2011.
- [19] ESO, *The M490 Mandate Smart Grids Reference Architecture*, European Standards Organizations, Nov. 2012.
- [20] M. Rohr, M. Specht, M. Stadler, M. Grndler, T. Luhmann, K. Wolf, N. Marwede, J. Wisch, and N. Vogel, "Realisierung einer smart grid-architektur," in *IT-Architekturentwicklung im Smart Grid*, H.-J. Apperath, P. Beenken, L. Bischofs, and M. Uslar, Eds. Springer Berlin Heidelberg, 2012, pp. 227–249.
- [21] J. Hughes *et al.*, "The integrated energy and communication systems architecture," *Electric Power Research Institute*, 2004.
- [22] C. Wietfeld, C. Muller, J. Schmutzler, S. Fries, A. Heidenreich, and H. Hof, "Ict reference architecture design based on requirements for future energy marketplaces," in *Smart Grid Communications (Smart-GridComm), 2010 First IEEE International Conference on*. IEEE, 2010, pp. 315–320.
- [23] R. Taylor, N. Medvidovic, and E. Dashofy, *Software architecture: foundations, theory, and practice*. Wiley Publishing, 2009.
- [24] T. Gruber *et al.*, "A translation approach to portable ontology specifications," *Knowledge acquisition*, vol. 5, no. 2, pp. 199–220, 1993.
- [25] D. Koß, D. Bytschkow, P. Gupta, B. Schatz, F. Sellmayr, and S. Bauereiß, "Establishing a smart grid node architecture and demonstrator in an office environment using the soa approach," in *Software Engineering for the Smart Grid (SE4SG), 2012 International Workshop on*. IEEE, 2012, pp. 8–14.
- [26] F. Arvidsson and A. Flycht-Eriksson, "Ontologies i," *PDF*. <http://www.ida.liu.se/~janma/SemWeb/Slides/ontologies1.pdf>. Retrieved, vol. 26, 2008.